BITS Pilani

Automotive Vehicle

08/08/2026

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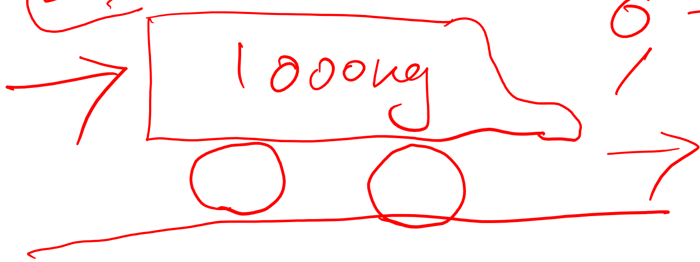
Department of Mechanical Engineering

Automotive Vehicle (AEL ZC441)
Lecture: Powertrain Fundamentals

Recap of Previous session

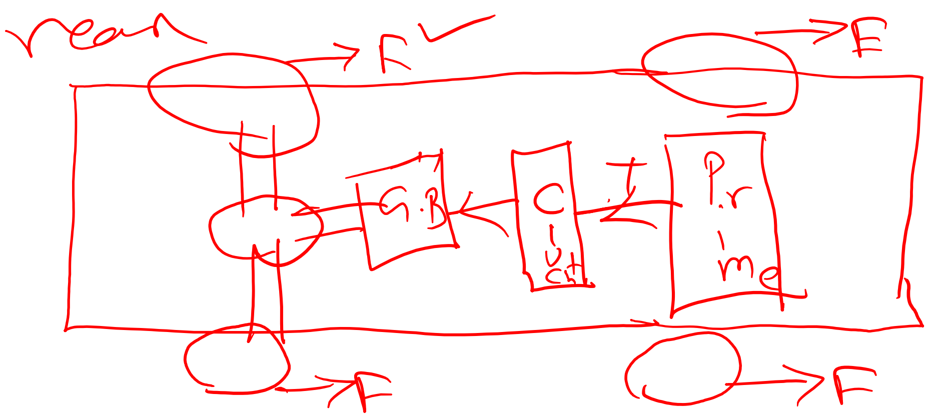
Why the prop. sys. is required?

5560 N ~ 56 Person



0 to 100 km per hour

5 sec



Front

Velocity

Top view

50 km/h

Key Powertrain Layouts

- FF - Front-engine, Front-wheel-drive
- FR - Front-engine, Rear-wheel-drive
- MR - Mid-engine, Rear-wheel-drive
- RR - Rear-engine, Rear-wheel-drive

Front-Engine, Front-Wheel-Drive (FF)

- **Layout**

Engine and transmission (transaxle) mounted transversely or longitudinally at the front, driving the front wheels only.

- **Applications**

The dominant layout for mainstream hatchbacks, sedans, and many crossovers, due to packaging efficiency and low cost.

- **Advantages**

Compact powertrain package, more interior space, good traction in slippery conditions due to weight over the driven wheels.

- **Disadvantages**

Front tyres must both steer and drive, limiting ultimate handling balance; tends toward understeer under power.

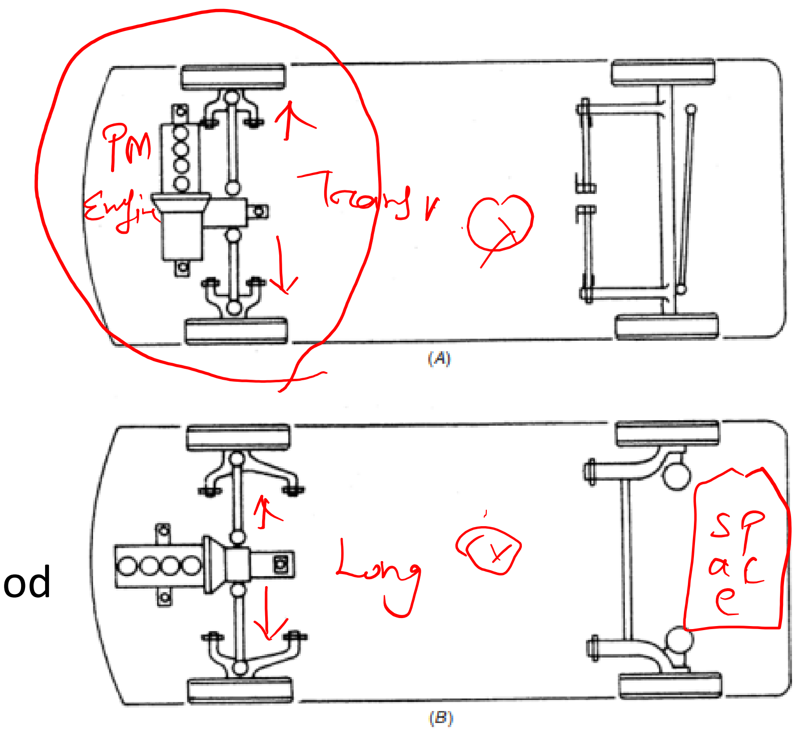


Fig. 11.25. Front-mounted engine and front-wheel drive.
A. Transverse front-mounted engine and front-wheel drive.
B. Longitudinal front-mounted engine and front-wheel drive.

Front-Engine, Rear-Wheel-Drive (FR)

- **Layout**

  Engine at the front, driving the rear wheels via a longitudinal transmission and propeller shaft.

- **Applications**

  Traditional layout for sedans, sports cars, and most trucks and buses.

- **Advantages**

  Separates steering (front) and driving (rear) duties between axles, giving a more balanced, tunable handling character.

- **Disadvantages**

  Requires a central transmission tunnel, reducing interior packaging efficiency versus FF.

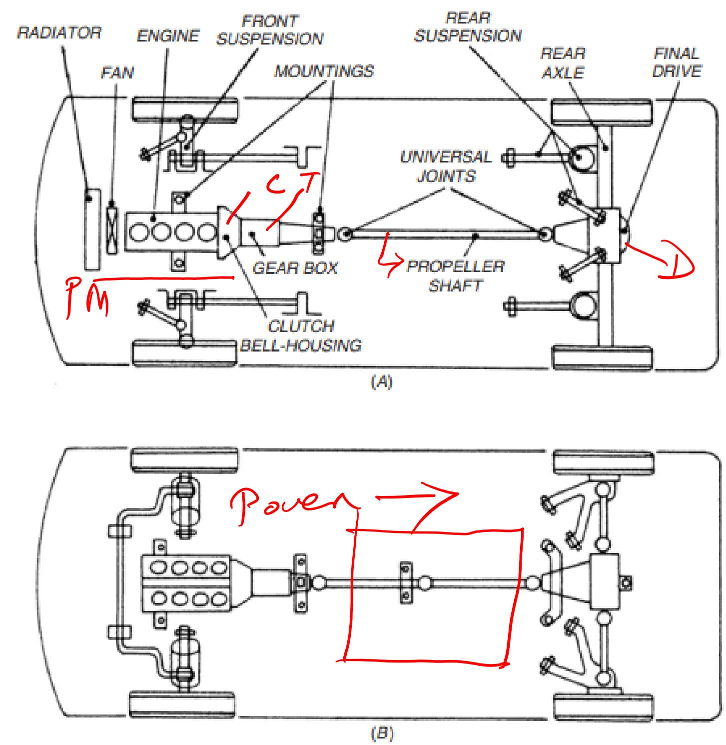


Fig. 11.24. Front-mounted engine and rear-wheel drive.
A. Un-sprung rigid rear axle and rear-wheel drive.
B. Independent rear suspensions and rear-wheel drive.

Mid-Engine, Rear-Wheel-Drive (MR)

- **Layout**

Engine mounted between the axles (typically behind the passenger compartment), driving the rear wheels.

- **Applications**

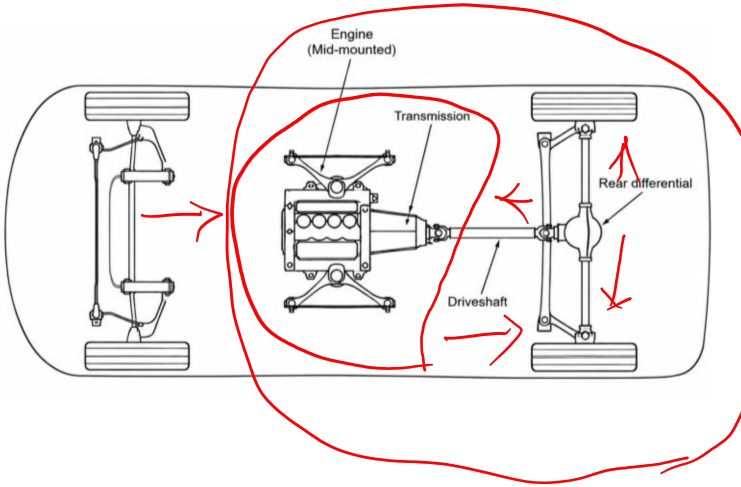
Common in sports cars and supercars where handling precision is prioritised over practicality.

- **Advantages**

Near-ideal weight distribution and giving very sharp, responsive handling.

- **Disadvantages**

Severely compromises passenger and cargo space - rarely suited to mainstream passenger vehicles.



Rear-Engine, Rear-Wheel-Drive (RR)

- **Layout**

Engine mounted behind the rear axle, driving the rear wheels.

- **Applications**

Historically significant (classic Volkswagen Beetle), still used today in some sports cars (Porsche 911).

- **Advantages**

Excellent rear-wheel traction under acceleration due to weight bias over the driven wheels.

- **Disadvantages**

Rear weight bias increases oversteer risk at the limit if not carefully tuned.

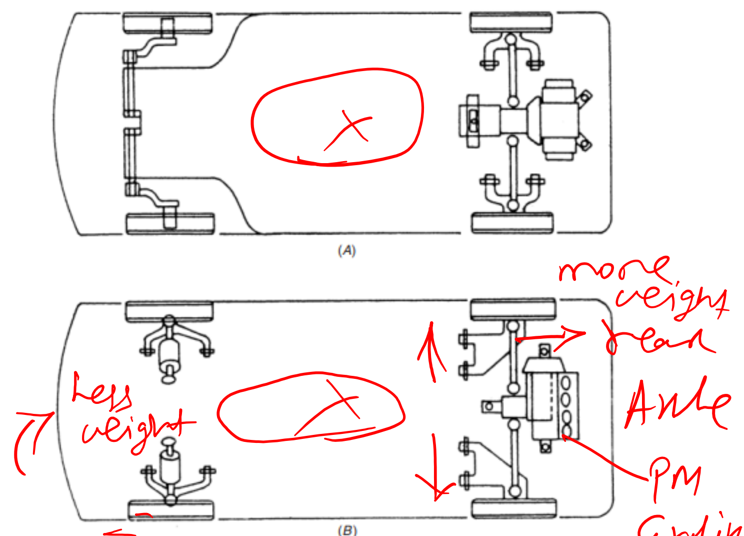
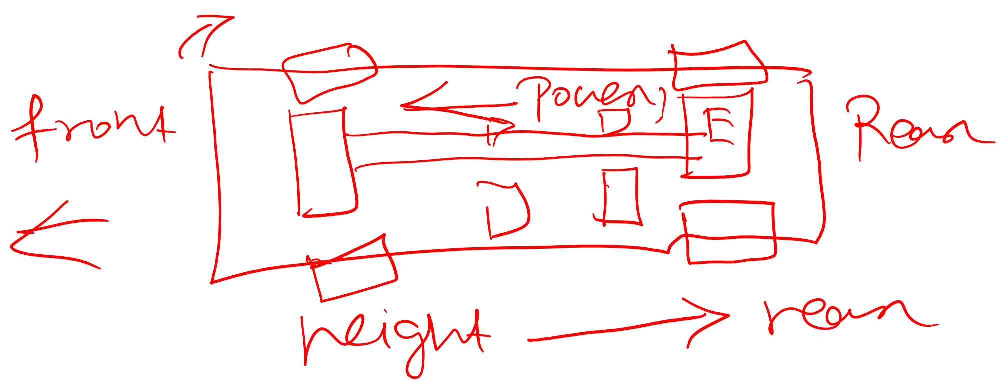


Fig. 11.26. Rear-mounted engine and rear-wheel drive.
A. Longitudinal rear-mounted engine and rear-wheel drive.
B. Transverse rear-mounted engine and rear-wheel drive.

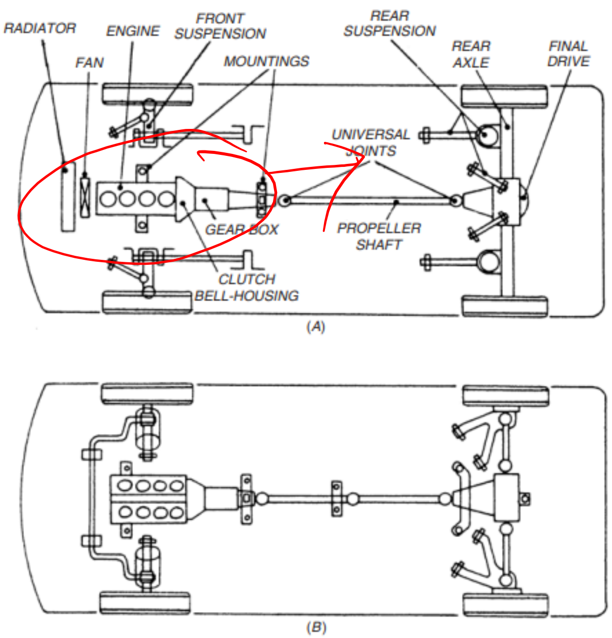
Rear-Engine, Front-Wheel-Drive (RF)

FF
RR
MR
FR
RF



Powertrain Layouts - Comparison

FERWD: Front Engine Rear Wheel Drive



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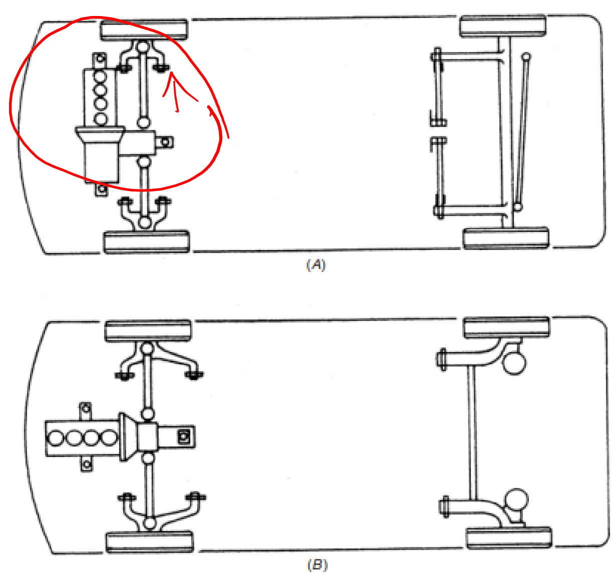

Fig. 11.25. Front-mounted engine and front-wheel drive.
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RERWD: Rear Engine Rear Wheel Drive

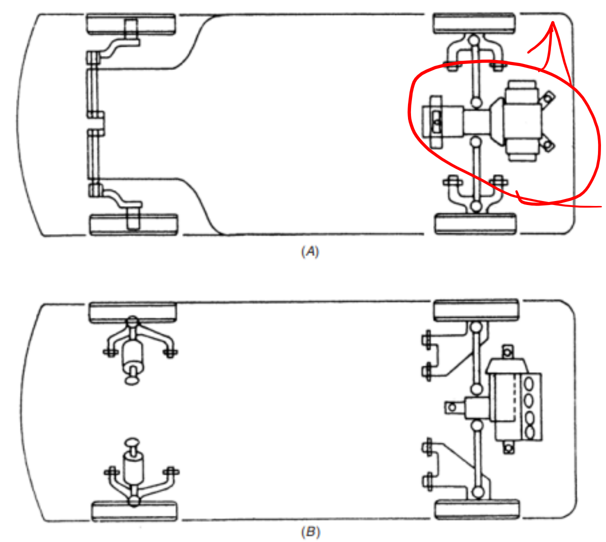


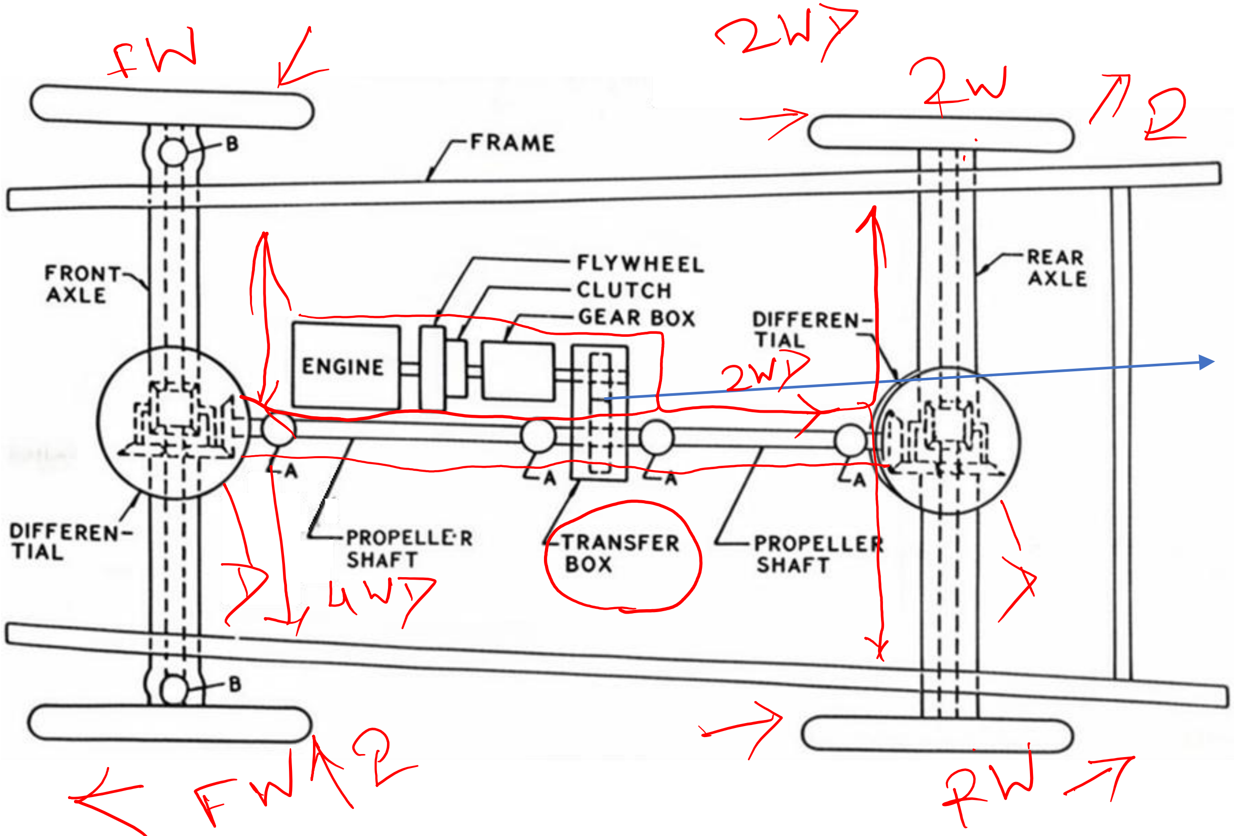
Fig. 11.26. Rear-mounted engine and rear-wheel drive.
A. Longitudinal rear-mounted engine and rear-wheel drive.
B. Transverse rear-mounted engine and rear-wheel drive.

FEFWD: Front Engine Front Wheel Drive

Powertrain Layouts - Comparison

Layout	Traction Character	Packaging
FF	Good in low grip (weight over driven wheels)	Most space-efficient
FR	Balanced	Requires transmission tunnel
MR	Excellent under acceleration	Poor - minimal cabin/cargo space
RR	Excellent under acceleration	Good cabin space, limited rear cargo

Four-Wheel-Drive (4WD)



Giving Power/to all the four wheels

Figure: Different components of four-wheel drive automobile (top view)

Four-Wheel-Drive (4WD)

- **Layout**

Power delivered to all four wheels, via a transfer case and front/rear differentials (or independent motors in an EV).

- **Applications**

Off-road vehicles, performance cars seeking maximum traction, and increasingly common across mainstream SUVs and EVs.

- **Advantages**

Mud, Snow

Best available traction and stability, particularly in low-grip conditions.

- **Disadvantages**

Added weight, cost, and driveline complexity versus a two-wheel-drive layout; typically a small efficiency penalty.

Two & Four wheel drive Vehicles

2WD

4WD

ECU Commands

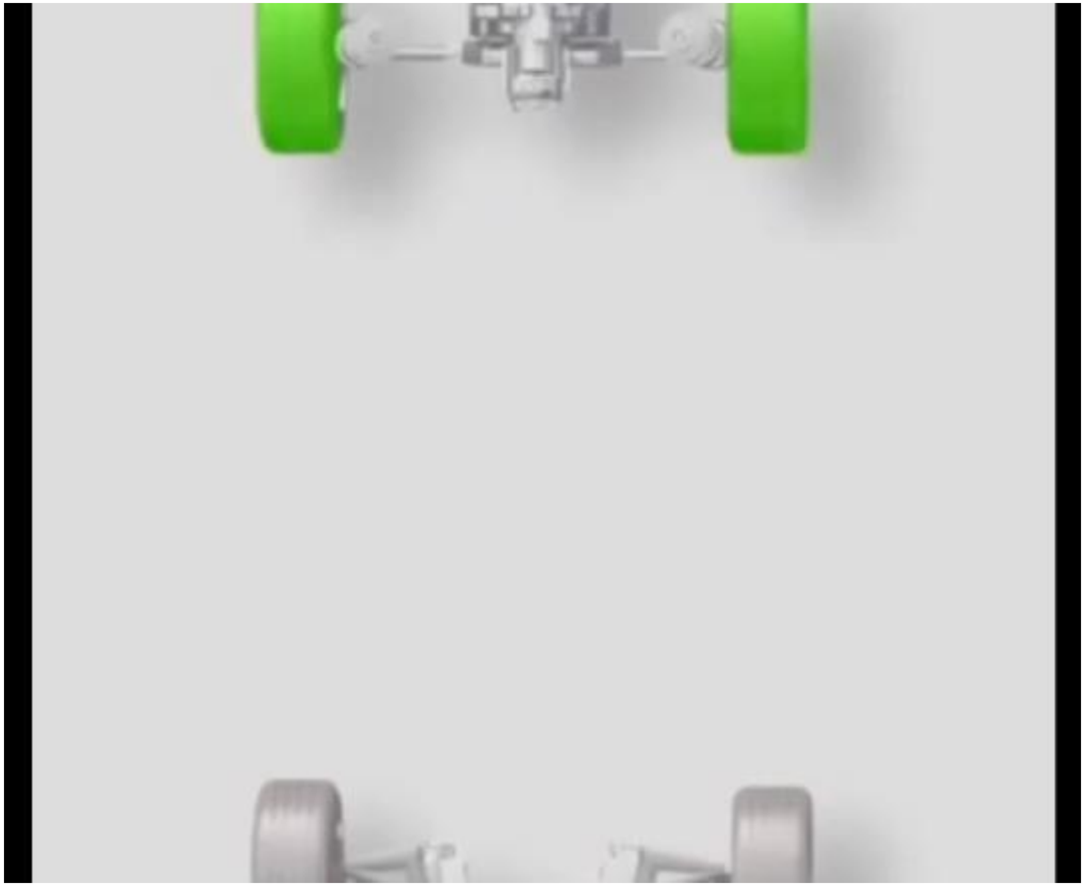
AWD → ALL wheel drive
Torque to all the wheels

4WD → driver selects
Torq/Power → 2 wheel
→ 4 wheels

Two & Four wheel drive Vehicles

Aspect	Two wheel drive (2WD)	Four wheel drive (4WD)
Traction	Struggles in snow, mud, and off-road	Excellent in in snow, mud, and off-road
Fuel Efficiency	Better as it is lighter and simpler drivetrain	Worse because of heavier components and drag
Cost	Cheaper to buy and maintain	More expensive upfront and in repairs
Weight	Lighter	Heavier (Transfer box, Drive shaft)
Best For	Daily commuting and in highways	Off-roading and snow/ice
Drawbacks	Limited grip in adverse conditions	Increased tyre wear

Comparison: FWD, RWD, AWD, 4WD



Tractive Force and Torque Requirement

3. Tractive Force and Torque Requirement

Forces Acting on a Wheel

- **Fx - Longitudinal force**

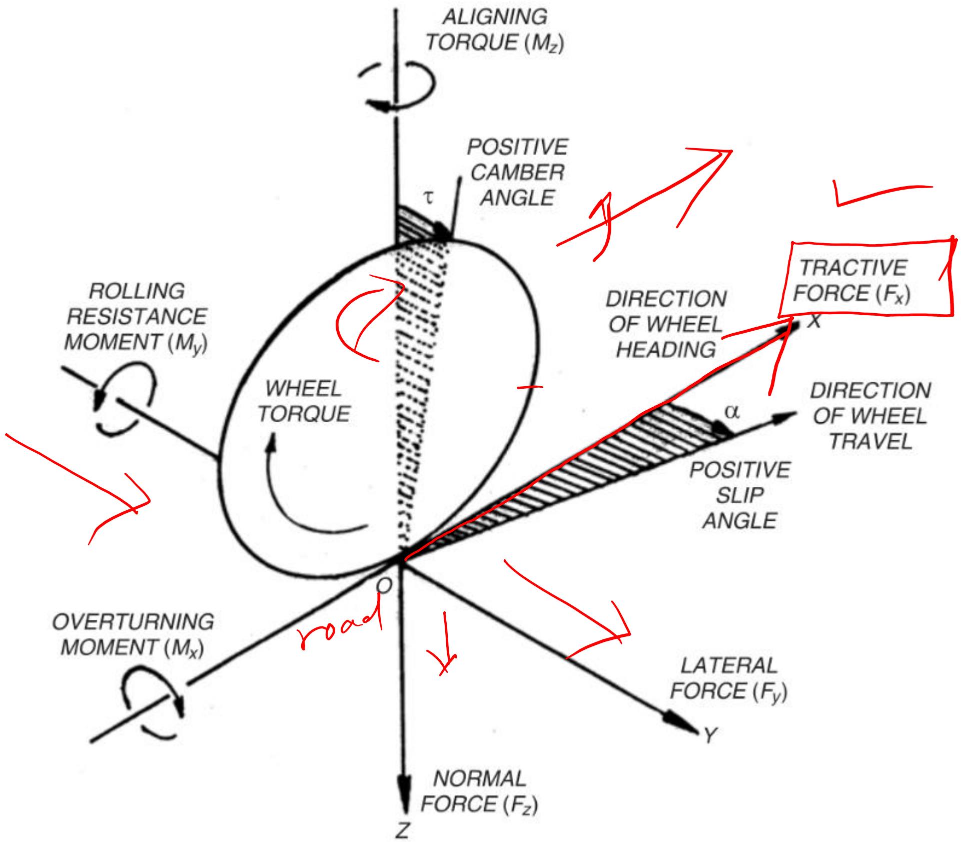
Acts along the direction of travel; produced when engine torque causes the wheel to rotate and push against the road (acceleration or braking).

- **Fy - Lateral force**

Acts sideways, generated when the vehicle corners and the tyre resists sliding sideways.

- **Fz - Normal (vertical) force**

The force pressing the tyre onto the road, arising directly from the vehicle's weight at that wheel.



What Is Tractive Effort?

- **Definition**

Tractive effort is the longitudinal force (Fx) available at the contact patch between the drive wheels and the road, produced when driving torque is applied.

- **Traction vs. tractive effort**

Traction is the tyre's ability to transmit this force to the road without slipping - usable tractive effort is always limited by available traction, not just engine output.

- **The design implication**

An engine can produce more torque than the tyres can usefully transmit - traction, not the powertrain, is often the true limiting factor at low speed.

Vehicle Motion Resistances - Overview

- **Aerodynamic (air) resistance**

Resistance from air being displaced as the vehicle moves - grows sharply with speed.

- **Rolling resistance**

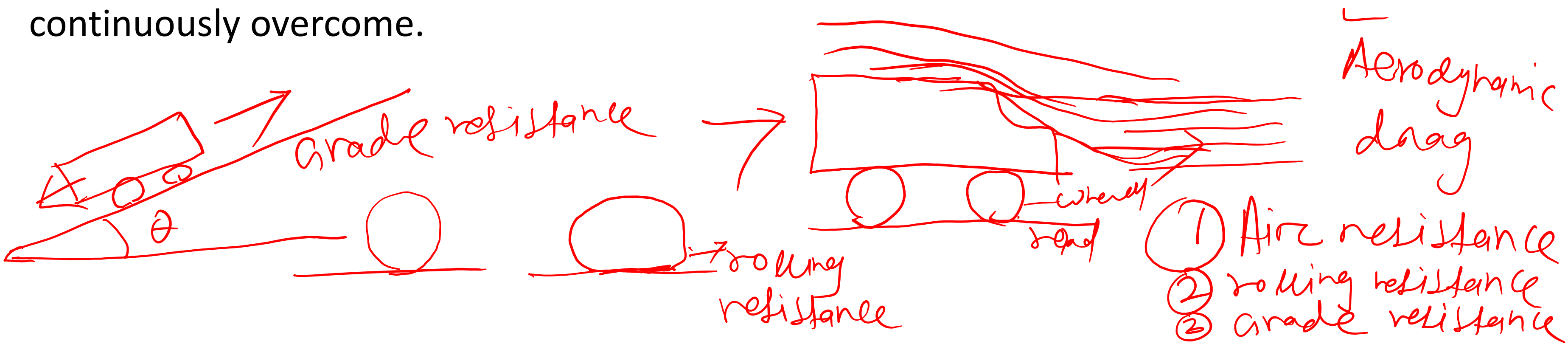
Resistance from tyre deformation as it rolls - present at all times the vehicle is moving.

- **Grade resistance**

The extra resistance from gravity when travelling up an incline - absent on level ground, significant on a slope.

- **Together**

These three resistances sum to the total road resistance the propulsion system must continuously overcome.



Aerodynamic (air) resistance - Causes

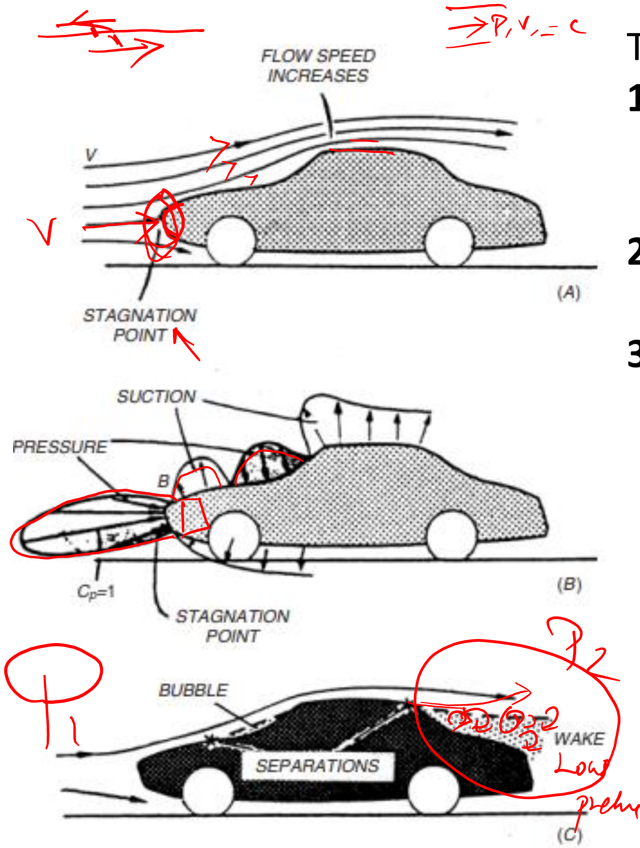


Fig. 11.21. Air flow and pressure distribution around a typical saloon car.

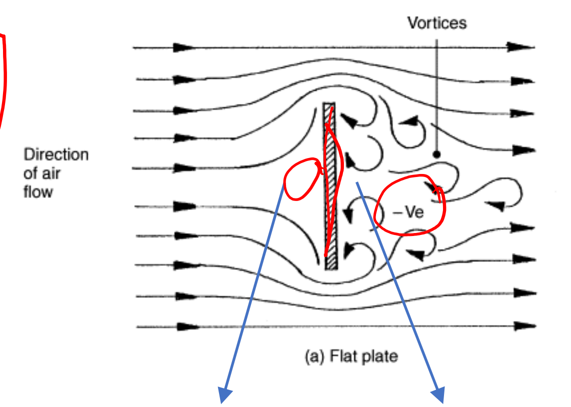
$\boxed{\Delta P = P_1 - P_2}$

Total aerodynamic resistance of a vehicle :

1. **Pressure Drag:** Caused by pressure difference between front and rear. Most significant portion of aerodynamic resistance (shape of the body).
2. **Skin Friction Drag:** Caused because of shear force exerted by the airstream on the vehicle surface.
3. **Internal Flow Resistance:** Caused by airflow through the radiator system, interior cooling passages, and the ventilation system.

$$R_a = \left(\frac{1}{2}\right) * C_d * \rho * A * V^2$$

R_a: Resistance air
C_d: Coefficient of discharge
ρ: Air density
A: Vehicle frontal area
V: Velocity of the vehicle



(a) Flat plate

High pressure region

Low pressure region

Air Resistance - Formula and Typical Coefficients

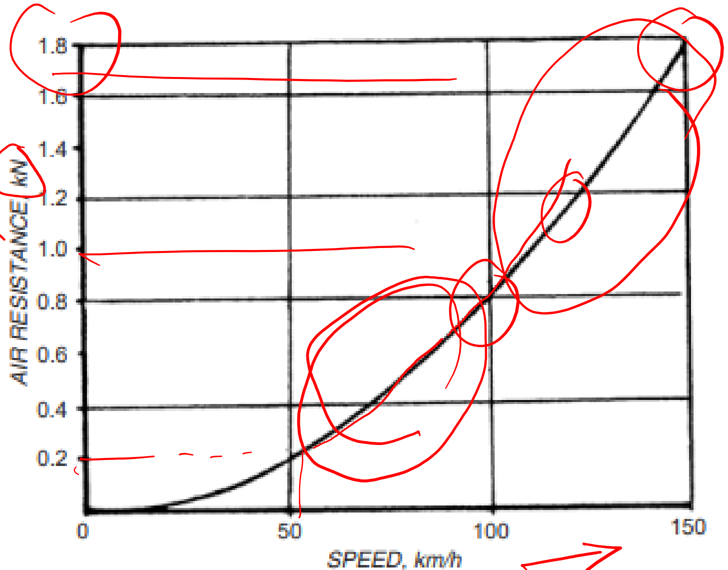


Fig. 11.22. Force required for overcoming air resistance.

$AR \propto V^2$

Table 11.2. Aerodynamic drag coefficient for different types of vehicles.

Types of Vehicle	*C_d (dimensionless)*
Racing Car	0.25 – 0.30
Passenger Car	0.30 – 0.60
Convertible	0.40 – 0.65
Bus	0.60 – 0.70
Truck	0.80 – 1.00
Tractor and Trailers	1.25 – 1.35
Motor Cycle	1.75 – 1.85

frontal Area

Air resistance is given by:

$$R_a = K_a A V^2 \qquad K_a = \frac{1}{2}\rho C_d$$

Low

Cd

Where:

A = projected frontal area (m²)

V = vehicle speed (km/h)

K_a = coefficient of air resistance

Rolling Resistance - Formula and Typical Coefficients

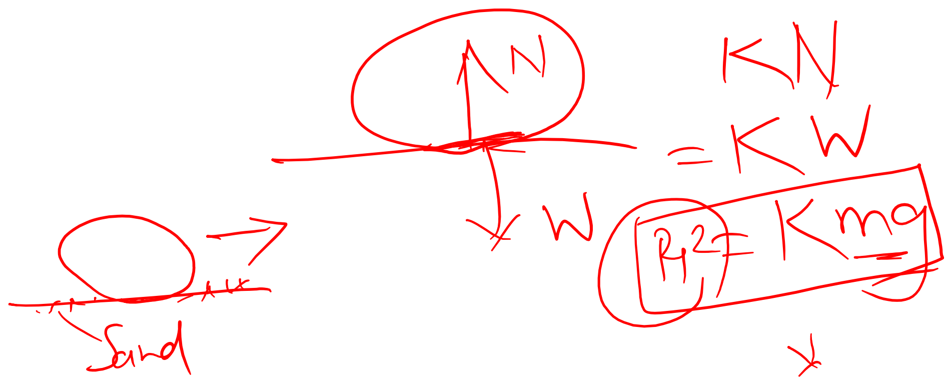
❖ The magnitude of rolling resistance depends on the following factors:

- Nature of the road surface
- Type of tyre (pneumatic or solid rubber)
- Weight of the vehicle
- Speed of the vehicle

❖ Rolling resistance is expressed as: KN = KW = Kmg

W = Total weight of the vehicle (N)
K = Rolling resistance coefficient (depends on road surface and tyre type)

Typical values of K:
0.0059 for good roads
0.18 for loose sand
0.015 as a representative value

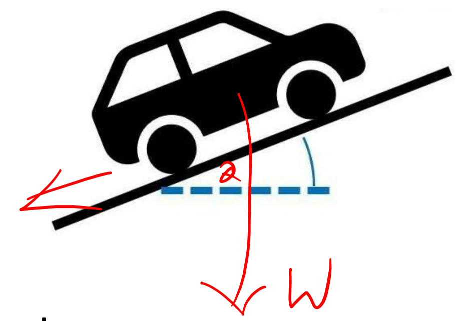


Grade Resistance

- Grade resistance is the component of vehicle weight acting along the slope of the road. It depends on the steepness of the gradient.
- If a gradient is given as 1 in 5, it means the vehicle rises 1 meter vertically for every 5 meters of horizontal travel.

Grade resistance is expressed as:

$$\boxed{R_g = W \sin\theta}$$



Where
W = total weight of the vehicle (N)
θ (theta) = angle of inclination of the slope

1m
5m

$\boxed{\tan\theta = \frac{1}{5}}$

Grade (%) = tanθ × 100

For small angles, tanθ ≈ sinθ

Small θ

$\boxed{\tan\theta = \sin\theta}$

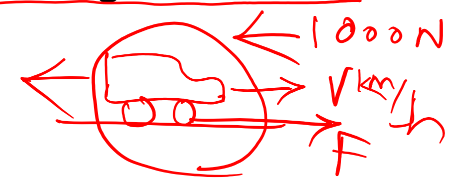
Total Road Resistance

- **Combining the three terms**

Total resistance, R = Rr + Ra + Rg (rolling, air, and grade resistances).

- **Why the total matters**

The propulsion system must be able to generate a tractive force greater than R at every operating condition the vehicle will face.



$\boxed{F = R}$

Total resistance

(Tractive force) $F > 1000$ N

1001 N

$$R_{air} + R_{rolling} + R_{grade} = 1000N$$

$$\frac{1}{2}\rho A C_d V^2 \qquad KW \qquad W\sin\theta$$

Tractive force

- Tractive effort is the force available at the contact between the drive wheels and the road. Traction is the ability of the tyres to transmit this force without slipping. The usable tractive effort is always limited by available traction.

η transmission

$$\boxed{\frac{60000\,P_E}{2\pi N}}$$

Force

Engine
N, ω

$P = T_E\,\omega$

$= \frac{2\pi N}{60}$

$P = \frac{2\pi N T_E}{60}$

$$\boxed{T_E = \frac{60 \times P \times 10^3}{2\pi N}}$$

Engine torque: $T_E = \dfrac{60000\,P_E}{2\pi N}$

Torque at Drive Wheels: $T_w = (g.r. \times a.r.)\,\eta_t T_E = \boxed{G\,\eta_t T_E}$

Tractive Effort: $F = \dfrac{T_w}{r} = \dfrac{T_E\,G\,\eta_t}{r}$

$F = T_w / r$

P_E = engine power (kW)

T_E = engine torque (Nm)

η_t = transmission efficiency

$g.r.$ = gearbox gear ratio

$a.r.$ = axle ratio

G = overall gear ratio

r = tyre radius (m)

N = engine speed (rpm)

If F>R (total resistance), the excess tractive effort is used for acceleration, hill climbing, and drawbar pull.

$R = R_{rr} + R_a + R_g$

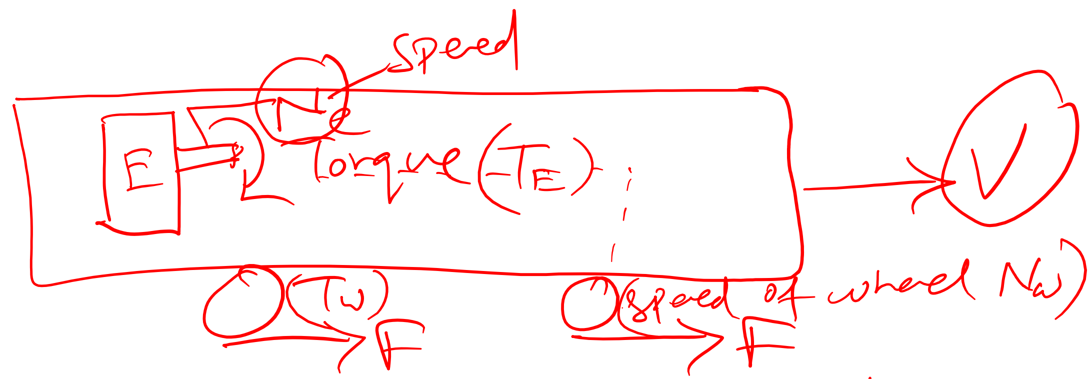
r

$F = \frac{T_w}{r}$ $\quad T_w = F \times r$ $\quad$ F

Engine speed - Vehicle speed

- Distance travelled by one wheel revolution = 2πr
- Wheel rpm = N/G [N: Engine Speed & G: Gear Ratio]
- Distance travelled by wheel in one minute (distance/min) = 2πr x (N/G)
- Convert to vehicle speed (m/s) V = 2πr x (N/60G)

$$\boxed{V = \frac{2\pi r N}{60 G}}$$

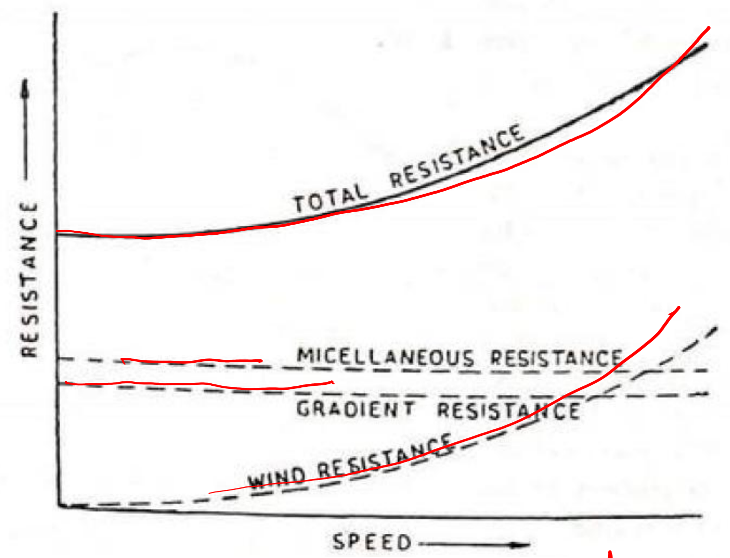


$2\pi r \times N/60G$ m/s

$N_w = \frac{N}{G}$

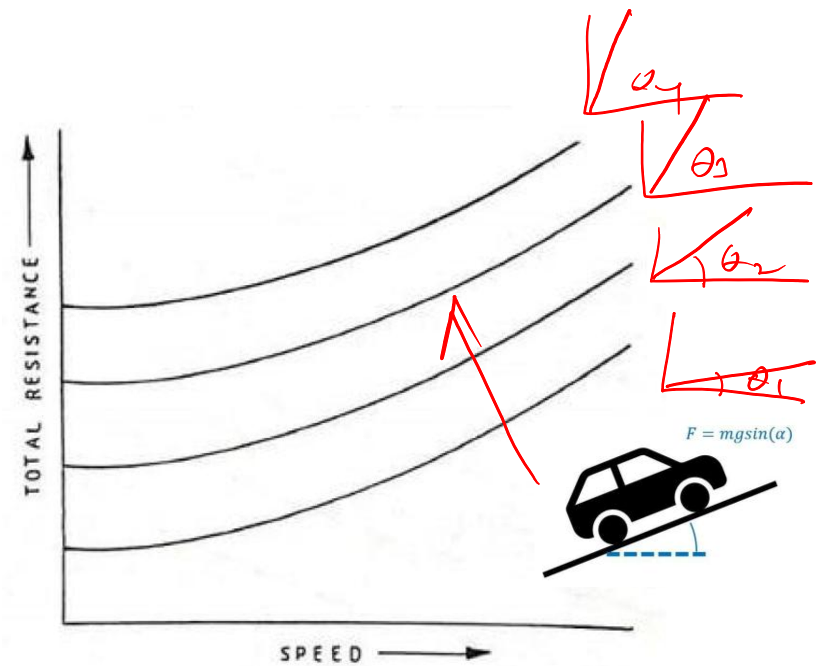
Total Resistances for Different Gradients

R



Speed

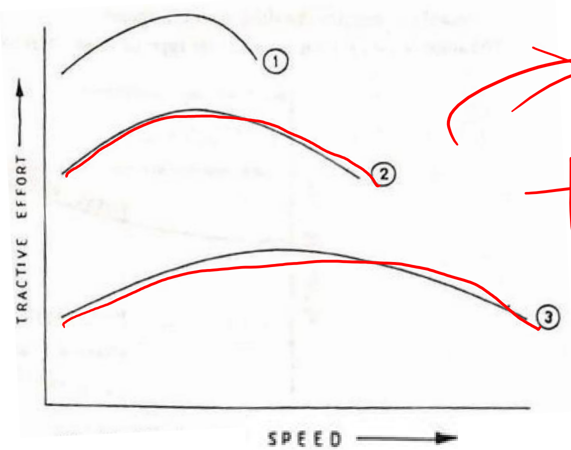
Various resistances to vehicle motion



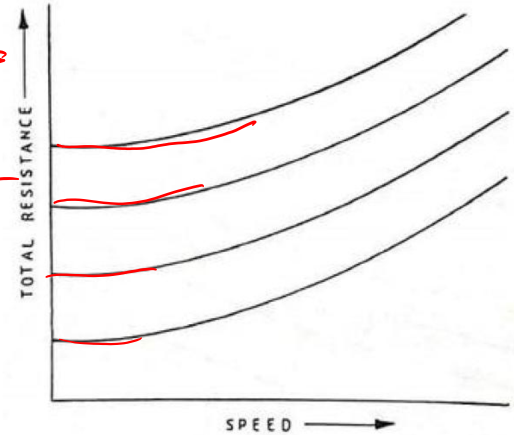
Total resistance for different gradients

Tractive Effort and Total Resistance

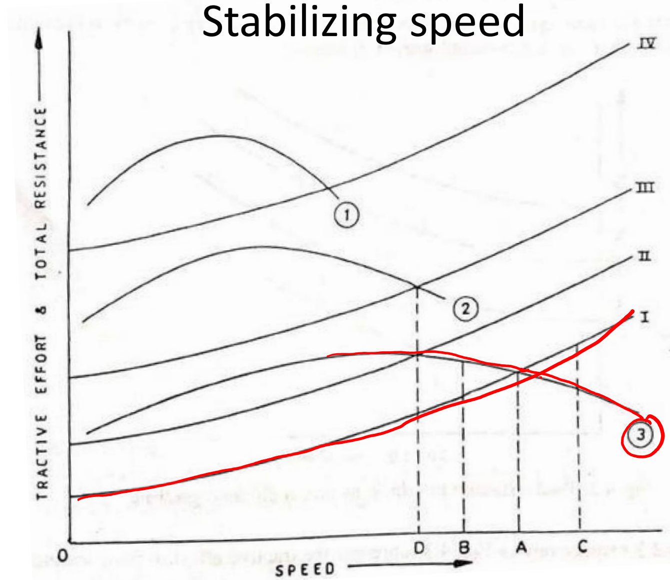
OA/OD-
Stabilizing speed



Tractive effort curves for different gears



Total resistance for different gradients



Superimposing tractive effort and resistance curves

Why you always prefer ?
1st gear in a very steep road or in hill, or in climbing a hill.

Vehicle stability at Different Gradients

1. OA represents the stabilizing speed on curve I.

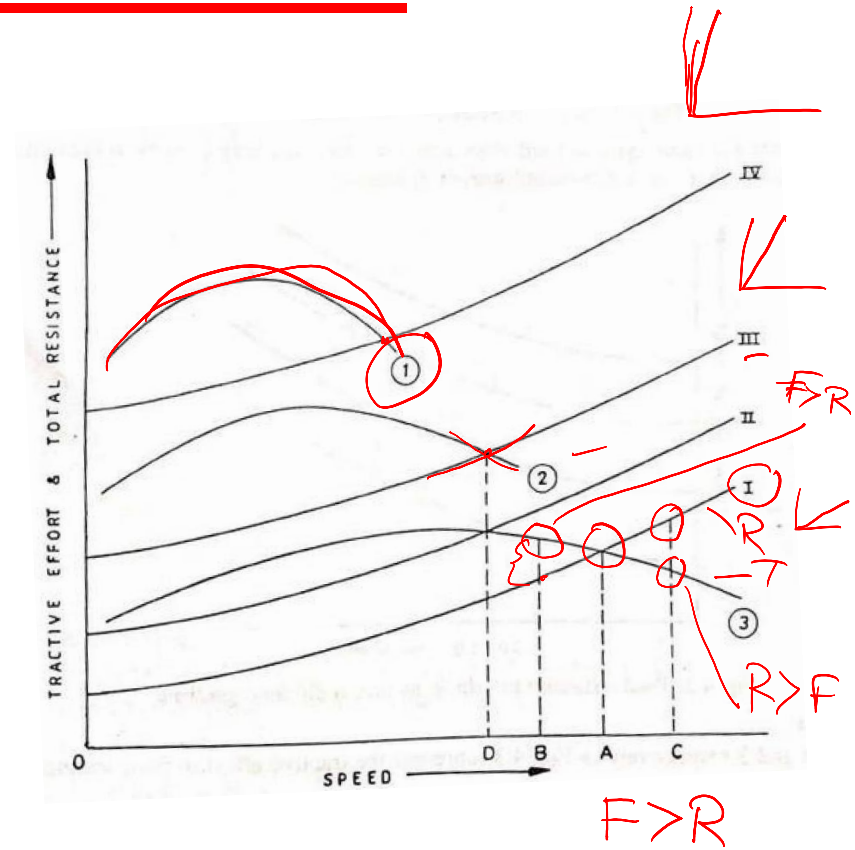
- If the vehicle speed is higher than OA (e.g., at OC), excess resistance causes deceleration down to OA.
- If the vehicle speed is lower than OA (e.g., at OB), excess tractive effort causes acceleration up to OA.

2. When the vehicle moves to the next gradient (II), the stabilizing speed decreases.

3. On a steeper gradient (III), the stabilizing speed reduces further.

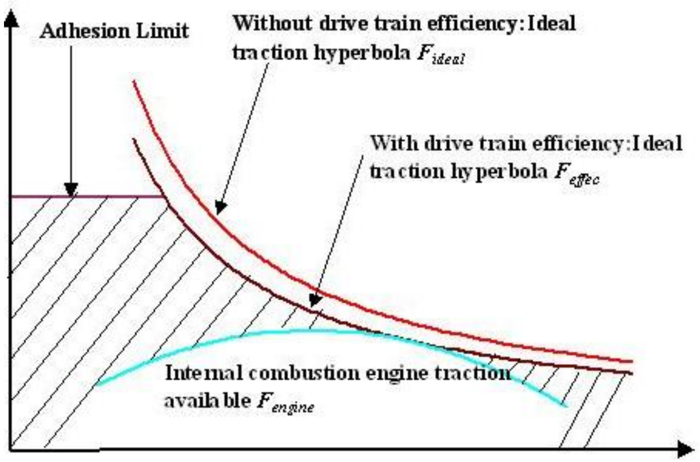
4. At this gradient, curve 3 nowhere intersect curve III, indicating that the vehicle cannot continue in top gear.

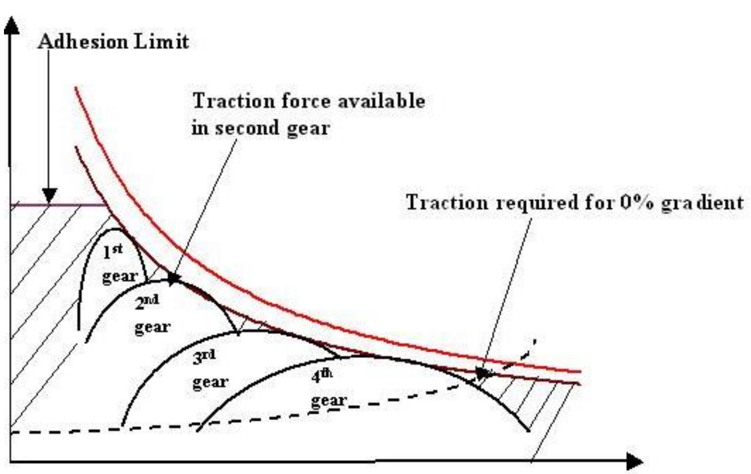
5. To maintain motion, the vehicle must be shifted to second gear, where a stabilizing speed OD is obtained.



Superimposing tractive effort and resistance curves

Gearbox: Why multi-speed





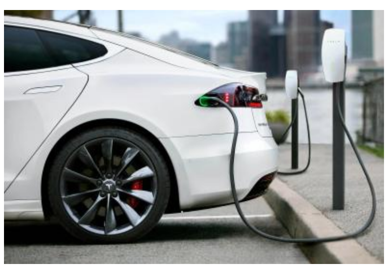
- ❖ **Tractive effort is the driving force available at the wheels that allows a vehicle to accelerate, climb, and overcome resistances**
- ❖ **Adhesion limit:** It is the maximum tractive (driving) force that can be transmitted between a vehicle's driven wheels and the road surface without wheel slip.

- When a wheel applies torque to the road, the force is transmitted through tire-road friction.
- This friction has a maximum value. Beyond this point, the tire loses grip and starts to slip or spin.
- That maximum usable force is called the adhesion limit.

Numerical 1

Design Battery capacity for a 400 km range EV?

Assume:
Speed: 100 km/h
Rolling res coef: 0.01
~~Air res coef: 0.28~~
Mass: 1000 kg
Frontal Area: 2.2 m2
Cd: 0.28



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200 k.m

200 k.m

400 k.m

Given

$V = 100\,km/h \approx 27.8$ (1000 / 3600)

$K_R = 0.01$

$C_d / K_a = 0.28$

$m = 1000\,kg$

$A = 2.2\,m^2$

$\boxed{F = 383.7\,N}$

Tractive Force > Resistance force

Total resistance =

$RF_{Rolling} + RF_{air} + RF_{grade} + F_{acc^n}$ ($RF_{grade} \to 0$, $F_{acc^n} \to 0$)

$= RF_R + RF_{air}$

$= KW + \frac{1}{2}\rho A C_d V^2$

$= (0.01 \times 1000 \times 9.81) + \left(\frac{1}{2} \times 1.2 \times 2.2 \times 0.2 \times 27.8^2\right)$

$= 98.1\,N + \cancel{285.6}\,N$ 291

$= \cancel{383.7}\,N$

389 N

Energy = F × distance = ~~383.7 × 1000 (m)~~

Energy

= 383.7 kJ = 383.7 / 3600

= kWh

Numerical 1

Design Battery capacity for a 400 km range EV?

Assume:
Speed: 100 km/h
Rolling res coef: 0.01
Air res coef: 0.28
Mass: 1000 kg
Frontal Area: 2.2 m2
Cd: 0.28

Vehicle mass (m) = 1000 kg
Average driving speed (v) = 100 km/h = 27.78 m/s
Rolling-resistance coefficient (Crr) = 0.01
Drag coefficient (Cd) = 0.28
Frontal area, A = 2.2 m²

Rolling Resistance Force:
Frr = Crr × m × g
Frr = 0.01 × 1000 × 9.81
Frr = 98.1 N

Aerodynamic drag Force:
Faero = ½ρCdAv²
Faero = ½ × 1.2 × 0.28 × 2.2 × (27.78)²
Faero = 291 N

Total Resistance Force:
Ftotal = 98.1 + 291

$$F_{\text{total}} = F_{\text{rolling}} + F_{\text{aero}} + \cancel{F_{\text{gradient}}}^{\,0} + \cancel{F_{\text{acceleration}}}^{\,0}$$

$\boxed{\textbf{Ftotal = 389.1 N}}$

Mechanical energy required for 1 km
Ewheel = Ftotal × distance
Ewheel = 389.1 × 1000
$\boxed{\text{Ewheel = 389100 J}}$

In kWh
Ewheel = 389100/3600
Ewheel = 108.1 Wh/km

$\boxed{\text{Ereal ≈ 150 Wh/km}}$

Energy required for 400 km
Eusable = 150 Wh/km × 400 km
Eusable = 60000 Wh
$\boxed{\textbf{Eusable = 60 kWh}}$

$= \frac{60}{0.8} = 75\,\text{kWh}$

90%
80%
70%

Numerical 2

Q2: An engine is required to power a truck having a gross weight of 40937 N. The maximum grade which the truck will have to negotiate at 32 km/h in second gear is expected to be 15%. The rolling resistance coefficient is 0.017 and the air resistance coefficient is 0.0324 in the following formula:

Rolling

→ 9.81 N

Total resistance = Kf W + Ka A V² (kgf) where, A = Frontal area (m²) V = Vehicle speed (km/h)

$\frac{1}{2}\rho C_d\} \rightarrow 0.0324$

The frontal area is 5.2 m². The transmission efficiency in second gear is estimated to be 80%. Calculate:

→ F × velocity
Traction force = Total resistance

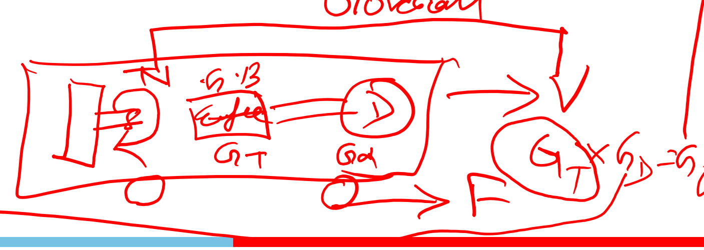
$V = 2\pi r \times \frac{N}{60}$

1.The minimum power which should be available from the engine.
2.The gear ratio in second gear if this power is available at 2400 rpm and the effective radius of the wheels is 0.419 m.

G =

3.The minimum speed of this vehicle in top gear on a level road at the same engine speed, assuming a transmission efficiency of 90% in top gear.
4.The gear ratio in top gear.

Gearbox



The differential has a reduction ratio of 3.92.

Solution: Q2

- Gross weight, $W = 40937$ N
- Speed in 2nd gear, $V = 32$ km/h
- Grade = 15%, So $\tan\theta = 0.15$ $\simeq \sin\theta$, $R_g = W \tan\theta$
- Rolling resistance coefficient, $K_f = 0.017$
- Air resistance coefficient, $K_a = 0.0324$
- Frontal area, $A = 5.2$ m^2
- Efficiency in 2nd gear, $\eta_2 = 0.80$
- Engine speed = 2400 rpm
- Wheel radius, $r = 0.419$ m
- Differential ratio = 3.92
- Top gear efficiency, $\eta_t = 0.90$

Solution: Q2

1. Grade Resistance

For small road angles

$\sin\theta = 0.15$

$R_g = W \sin\theta$

$R_g = 40937 \times 0.15$

$R_g = 6073$ N

2. Rolling Resistance

$R_r = K_r W$

$R_r = 0.017 \times 40937$

$R_r = 696$ N

3. Air Resistance

$R_a = K_a A V^2$

Substituting,

$R_a = 0.0324 \times 5.2 \times 32^2$

$R_a = 172.5$ kgf
$= 1692$ N

32 km/h

4. Total resistance:

$R = R_r + R_a + R_g$

$R = 696 + 1692 + 6075$

$R = 8461$ N

5. Power required from engine: F × V

$BP = RV / (\eta_t \times 1000)$

Here speed must be in m/s, so:

$V = 32 / 3.6$ m/s

Thus,

$BP = [8461 \times (32 / 3.6)] / (0.8 \times 1000)$

$BP = 94$ kW

6. Overall gear reduction in second gear

$V = 2\pi N r / 60G$ m/min

$G = 2\pi N r / 60V$ m/s

Substituting values:

$G = (2\pi \times 2400 \times 0.419 \times 3.6) / (60 \times 32)$

$G = 11.85$

7. Calculate second gear ratio

Overall reduction = gearbox ratio × differential ratio

Second gear ratio = 11.85 / 3.92 = 3.02:1

8. Maximum speed in top gear on level road

Rg = 0 Resistance = Rr + Ra

So total resistance becomes:

$R = R_r + R_a = 696 + 1.65V^2$

$R = (BP \times \eta_t \times 3600) / V$

$696 + (0.1685 \times V^2 \times 9.81) = (94 \times 0.9 \times 3600) / V$

$V = 54.5$ km/h

10. overall reduction in top gear

$G = 2\pi N r / V$

$G = (2\pi \times 2400 \times 0.419 \times 3.6) / (54.5 \times 1000)$

$G = 6.96$

11. Top Gear ratio: 6.96 / 3.92 = 1.78:1

Summary

- Vehicle propulsion requirements are set by three resistances - rolling, aerodynamic, and grade - which the powertrain must overcome at every speed.
- Tractive force and torque translate engine output, through gearing, into the usable force at the tyre contact patch - always capped by available traction.
- Powertrain architecture determines how that torque path is laid out across the vehicle, shaping handling, packaging, and traction together.